

PSP AWS switchover to US-EAST (ReadWrite) tip-toe steps

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Pre-requisites

- Verify latest veridata compare run to make sure there are NO OOS rows for IBOB + Monolith
 - In case of OOS, follow steps as per belowexample

```
1. log in to db as GGT user

2. run 15 days veridata compare check OOS rows in veridata for table and OOS rows and columns

3. run below update on target(12c) database as GGT user. Change table name as per OOS report.
update pspadm.PSP_ADDRESS set (ADDRESS_LINE2)=(select ADDRESS_LINE2 from pspadm.PSP_ADDRESS@to_source
where ADDRESS_SEQ='bd8ae74e-d3a4-4fb9-a2c4-eb0cf9c2c7b6' and REALM_ID='-1')
where ADDRESS_SEQ='bd8ae74e-d3a4-4fb9-a2c4-eb0cf9c2c7b6' and REALM_ID='-1';
```

- us-west & us-east - Ensure Golden gate replication is working between two

1. us-west - monolith

GGSCI (ggpsppp001-a.sbg-ppsp-prod.a.intuit.com) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSP001	00:00:01	00:00:05
EXTRACT	RUNNING	PPSP001	00:00:00	00:00:08
REPLICAT	RUNNING	RSPS001	00:00:00	00:00:01

us-east - monolith:

GGSCI (ggpspsp001-a.sbg-ppsp-prod.a.intuit.com) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSP001	00:00:04	00:00:08
EXTRACT	RUNNING	PPSP001	00:00:00	00:00:02
REPLICAT	RUNNING	RSPSP001	00:00:06	00:00:02

us-west - IBOB

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EIBPP001	00:00:03	00:00:05
EXTRACT	RUNNING	PIBQPD03	00:00:00	00:00:05
EXTRACT	RUNNING	PIBSP001	00:00:00	00:00:02
REPLICAT	RUNNING	RIBQPD03	00:00:00	00:00:00
REPLICAT	RUNNING	RIBSP001	00:00:18	00:00:08

us-east - IBOB:

GGSCI (ggpspspib1-a.sbg-ppsp-prod.a.intuit.com) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EIBSP001	00:00:04	00:00:00
EXTRACT	RUNNING	PIBPP001	00:00:00	00:00:01
REPLICAT	RUNNING	RIBPP001	00:00:07	00:00:08

2. Check lag via heartbeat on Monolith and IBOB

login to us-east

Monolith :

```
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from pspadm.gg_heartbeat;
```

IBOB :

```
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from ibobadm.gg_heartbeat;
```

3. Verify critical table data sanity

Monolith :

From 09/01 to 10/31 - select count(1) from pspadm.psp_financial_transaction partition(FINANCIAL_TXN_MG52019);

IBOB :

From 10/01 to 10/31 - select count(1) from ibobadm.PSP_SOURCE_SYSTEM_TRANSMISSION partition(SRCSYSTRNS_M102019);
From 11/01 to 11/31 - select count(1) from ibobadm.PSP_SOURCE_SYSTEM_TRANSMISSION partition(SRCSYSTRNS_M112019);

- AWS Monolith - Ensure all user/app accounts are locked except below ones :
 - PSPREAD_AWS, PSPBOADMIN, GGS, GGT, RDSADMIN, RDS_WAVEFRONT, SBG_WAVEFRONT
- AWS IBOB - Ensure all user/app accounts are locked except below ones :
 - IBOB_PSPREAD_AWS, GGS, GGT, RDSADMIN, RDS_WAVEFRONT, SBG_WAVEFRONT

- Check and compare synonyms for PSPAPP and PSPREAD user in us-west and us-east

```
select owner,SYNONYM_NAME from dba_synonyms where owner='PSPAPP' minus select owner,SYNONYM_NAME from dba_synonyms@T0_UE2 where owner='PSPAPP';
```

- AWS - Test and ensure cloud watch page alerts are working
- Review IOPS and Instance class settings for monolith and IBOB RDS.

Tip-toe steps

- Lock application user accounts and keep few readonly user unlocked on us-west database.

```
1.
Monolith :
alter user PSPADM account lock;
alter user PSPAPP account lock;

IBOB :
alter user IBOBADM account lock;
alter user IBOB_PROD_PSPAPP account lock;
alter user IBOB_PROD_PSPREAD account lock;

2.Monolith:
Kill existing connections (if still exist)
select 'exec rdsadmin.rdsadmin_util.kill('||s.sid||','||s.serial#||','||''IMMEDIATE''||');' from gv$session
s where username in ('PSPADM','PSPAPP') ;

IBOB:
Kill existing connections (if still exist)
select 'exec rdsadmin.rdsadmin_util.kill('||s.sid||','||s.serial#||','||''IMMEDIATE''||');' from gv$session
s where username in ('IBOBADM','IBOB_PROD_PSPAPP') ;
```

- Verify account_status is LOCKED for all below users

```
Monolith:
select username,account_status from dba_users where username in ('PSPADM','PSPAPP');

/* This is not required for Tip-toe #4
select username,account_status from dba_users where username in
('PSPADM','PSPAPP','INTU_IQUATE','OIMUSER','PSPREAD','PSPLOG','PSPREPORTS','APPBUGFIX','PSPBOADMIN');
*/

IBOB:
select username,account_status from dba_users where username like '%IBOB%';
```

- Verify GG replication

1. us-west - monolith

GGSCI (ggpsppp001-a.sbg-ppsp-prod.a.intuit.com) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSP001	00:00:01	00:00:05
EXTRACT	RUNNING	PPSP001	00:00:00	00:00:08
REPLICAT	RUNNING	RSPS001	00:00:00	00:00:01

us-east - monolith:

GGSCI (ggpspsp001-a.sbg-ppsp-prod.a.intuit.com) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSP001	00:00:04	00:00:08
EXTRACT	RUNNING	PPSP001	00:00:00	00:00:02
REPLICAT	RUNNING	RSPSP001	00:00:06	00:00:02

us-west - IB0B

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EIBPP001	00:00:03	00:00:05
EXTRACT	RUNNING	PIBQPD03	00:00:00	00:00:05
EXTRACT	RUNNING	PIBSP001	00:00:00	00:00:02
REPLICAT	RUNNING	RIBQPD03	00:00:00	00:00:00
REPLICAT	RUNNING	RIBSP001	00:00:18	00:00:08

us-east - IB0B:

GGSCI (ggpspspib1-a.sbg-ppsp-prod.a.intuit.com) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EIBSP001	00:00:04	00:00:00
EXTRACT	RUNNING	PIBPP001	00:00:00	00:00:01
REPLICAT	RUNNING	RIBPP001	00:00:07	00:00:08

- Wait for 5 mins

1. Check lag via heartbeat on Monolith and IB0B

Monolith :

login to us-west

```
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from pspadm.gg_heartbeat;
```

IB0B :

login to pprdpspdb903.ie.intuit.net

```
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from ibobadm.gg_heartbeat;
```

- restart extract EPSP001 on us-west
- check and verify sequence between us-west and us-east

```

set pagesize 1000
set numwidth 30

col SEQUENCE_OWNER format a10
col SEQUENCE_NAME format a32

select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER in ('PSPADM') order
by SEQUENCE_OWNER, SEQUENCE_NAME;

```

- Unlock application user accounts on AWS

```

1.
Monolith :
alter user PSPADM account unlock;
alter user PSPAPP account unlock;

IBOB:
alter user ibobadm account unlock;
alter user IBOB_PROD_PSPAPP account unlock;
alter user IBOB_PROD_PSPREAD account unlock;

```

- verify accounts are unlocked

```

Monolith:
select username,account_status from dba_users where username in ('PSPADM','PSPAPP');

IBOB:
select username,account_status from dba_users where username like '%IBOB%';

```

- Verify sequence replication between us-west and us-east (one time step)

```

set pagesize 1000
set numwidth 30

col SEQUENCE_OWNER format a10
col SEQUENCE_NAME format a32

select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER in ('PSPADM') order
by SEQUENCE_OWNER, SEQUENCE_NAME;

```

Post tip-toe steps

- Verify GG sync and ensure no lag between us-west and us-east
- On AWS

```
stats EPSPPP01
```

Rollback

- stop application and check for connections. Kill if any

```
Monolith:
Kill existing connections (if still exist)
select  'exec rdsadmin.rdsadmin_util.kill('||s.sid||','||s.serial#||','||''IMMEDIATE''||');' from gv$session
s where username in ('PSPADM','PSPAPP') ;

IBOB:
Kill existing connections (if still exist)
select  'exec rdsadmin.rdsadmin_util.kill('||s.sid||','||s.serial#||','||''IMMEDIATE''||');' from gv$session
s where username in ('IBOBADM','IBOB_PROD_PSPAPP') ;
```

- Lock application user accounts on AWS database.

1.

```
Monolith:
alter user PSPADM account lock;
alter user PSPAPP account lock;

IBOB:
alter user IBOBADM account lock;
alter user IBOB_PROD_PSPREAD account lock;
alter user IBOB_PROD_PSPAPP account lock;
```

- verify accounts are locked

1.

```
Monolith:
select username,account_status from dba_users where username in ('PSPADM','PSPAPP');

IBOB
select username,account_status from dba_users where username like '%IBOB%';
```

- Verify GG replication

```
1. us-west - monolith
GGSCI (ggpspp001-a.sbg-psp-prod.a.intuit.com) 1> info all

Program      Status      Group      Lag at Chkpt  Time Since Chkpt
MANAGER      RUNNING
EXTRACT      RUNNING     EPSP001    00:00:01      00:00:05
EXTRACT      RUNNING     PPSP001    00:00:00      00:00:08
REPLICAT     RUNNING     RSPS001    00:00:00      00:00:01

us-east - monolith:
GGSCI (ggpspp001-a.sbg-psp-prod.a.intuit.com) 1> info all
```

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EPSPS001	00:00:04	00:00:08
EXTRACT	RUNNING	PPSPP001	00:00:00	00:00:02
REPLICAT	RUNNING	RPSP001	00:00:06	00:00:02

us-west - IBOB

GGSCI (ggpsppib1.sbg-ppsp-prod.a.intuit.com) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
MANAGER	RUNNING			
EXTRACT	RUNNING	EIBPP001	00:00:03	00:00:05
EXTRACT	RUNNING	PIBQPD03	00:00:00	00:00:05
EXTRACT	RUNNING	PIBSP001	00:00:00	00:00:02
REPLICAT	RUNNING	RIBQPD03	00:00:00	00:00:00
REPLICAT	RUNNING	RIBSP001	00:00:18	00:00:08

us-east - IBOB:

GGSCI (ggpsppib1-a.sbg-ppsp-prod.a.intuit.com) 1> info all

Program	Status	Group	Lag at Chkpt	Time Since Chkpt
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EXTRACT	RUNNING	EIBSP001	00:00:04	00:00:00
EXTRACT	RUNNING	PIBPP001	00:00:00	00:00:01
REPLICAT	RUNNING	RIBPP001	00:00:07	00:00:08

- Wait for 5 mins

1. Check lag via heartbeat on Monolith and IBOB

login to us-west

Monolith :

```
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from pspadm.gg_heartbeat;
```

IBOB :

```
ALTER SESSION SET NLS_DATE_FORMAT='mm-dd-yyyy hh24:mi:ss';
select * from ibobadm.gg_heartbeat;
```

- restart extract EPSPS001 on us-east
- check and verify sequence between us-west and us-east

```
set pagesize 1000
set numwidth 30
```

```
col SEQUENCE_OWNER format a10
col SEQUENCE_NAME format a32
```

```
select SEQUENCE_OWNER, SEQUENCE_NAME, LAST_NUMBER from dba_sequences where SEQUENCE_OWNER in ('PSPADM') order
by SEQUENCE_OWNER, SEQUENCE_NAME;
```

- Unlock application user accounts on IHP

1.

Monolith :

```
alter user PSPADM account unlock;
```

```
alter user PSPAPP account unlock;

IBOB:
alter user ibobadm account unlock;
alter user IBOB_PROD_PSPAPP account unlock;
alter user IBOB_PROD_PSPREAD account unlock;
```

- verify accounts are unlocked

```
Monolith:
select username,account_status from dba_users where username in ('PSPADM','PSPAPP');

IBOB
select username,account_status from dba_users where username like '%IBOB%';
```

- For reference ONLY

```
/* For reference. Not required for tip toe #2, tip toe#3, tip toe#4
-- Unlock below ONLY IF locked in IHP
alter user PSPBOADMIN account unlock;
alter user PSPFLUXADM account unlock;
alter user PSPLOG account unlock;
alter user PSPFLUXADMIN08 account unlock;
alter user PSPLOG account unlock;
alter user PSPREAD account unlock;
alter user PSPTEMP account unlock;
alter user APPBUGFIX account unlock;
alter user ARCADM account unlock;
alter user EBIETL account unlock;
alter user ICADMIN account unlock;

2. verify
select username,account_status from dba_users where username like '%PSP%' or username in
('APPBUGFIX','ARCADM','ICADMIN','EBIETL');
USERNAME                                ACCOUNT_STATUS
-----                                -
ARCADM                                OPEN
PSPHIST_QBOE                           OPEN
APPBUGFIX                              OPEN
PSPLOG                                 OPEN
PSPFLUXADM                             OPEN
PSPFLUXADMIN08                         OPEN
PSPADM                                 OPEN
PSPTEMP                                OPEN
PSPBOADMIN                             OPEN
PSPAPP                                 OPEN
PSPREPORTS                             OPEN
EBIETL                                 OPEN
PSPREAD                                OPEN
ICADMIN                                OPEN

14 rows selected.
```

Post rollback steps

- Verify GG sync and ensure no lag between AWS and IHP
- On IHP

Performance tuning

Copy SQL Plan - IHP to AWS

- from cursor cache

```
oracle@pprdpspdb900.ie.intuit.net:/home/oracle/kpopat->./create_spb_aws.sh 627bwsf9sj3g4 803259958 PSP PSP-UW2

Copying plan for sql_id 627bwsf9sj3g4 from cluster# PSP to cluster# PSP-UW2 for plan hash value 803259958

Source DBL:SPM_PSP_LAS
Target DBL:SPM_PSP_UW2
Exporting sql plan for sqlid [627bwsf9sj3g4]
SQL plan export complated!
Importing sql plan for sqlid [627bwsf9sj3g4]
SQL plan import complated!
Export/Import of sql plan completed for sqlid [627bwsf9sj3g4] plan hash [803259958]

PL/SQL procedure successfully completed.
```

- from AWR repository

```
oracle@pprdpspdb900.ie.intuit.net:/home/oracle/kpopat->./create_spb_aws_awr.sh 1wbsa04w60gzt 3307958781 PSP PSP-UW2 4085 4086

Copying plan for sql_id 1wbsa04w60gzt from cluster# PSP to cluster# PSP-UW2 for plan hash value 3307958781

Source DBL:SPM_PSP_LAS
Target DBL:SPM_PSP_UW2
Exporting sql plan for sqlid [1wbsa04w60gzt]
SQL plan export complated!
Importing sql plan for sqlid [1wbsa04w60gzt]
SQL plan import complated!
Export/Import of sql plan completed for sqlid [1wbsa04w60gzt] plan hash [3307958781]

PL/SQL procedure successfully completed.
```